

Master's Thesis

Variance-Based Analysis of Predictive Multiplicity in Rashomon Sets

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Abstract

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1 Introduction

Supervised machine learning (ML) models are widely used to support decision making in areas such as healthcare, finance, and criminal justice. Their appeal lies in strong predictive performance, but this often comes at the cost of transparency: modern ML pipelines typically yield not a single model, but a collection of distinct models that achieve nearly identical validation loss. These models can nevertheless disagree on their predictions for particular individuals. In high-stakes settings, this means outcomes may depend on arbitrary choices such as model family, hyperparameters, and random seeds rather than on the data alone.

This thesis studies predictive multiplicity through Rashomon sets of near-optimal models, with a focus on an observation-level question: for which individuals are predictions unstable across equally good models, and does this instability exhibit structure in feature space? To answer this, we build a pipeline that (i) trains diverse models across multiple families and hyperparameter configurations, (ii) constructs a Rashomon set using a tolerance parameter ε , (iii) computes observation-wise predictive variance across the Rashomon set, and (iv) analyzes whether high-variance observations form coherent regions in feature space using k -nearest-neighbor graphs and spatial statistics.

1.1 Motivation: a concrete audit scenario

Consider a bank that uses a probabilistic classifier to support credit decisions. Each applicant receives a predicted probability of default, which is then mapped to an action such as approve, manual review, or decline. For model governance, the bank periodically audits the stability of its system: rather than evaluating only a single deployed model, the audit examines the set of near-optimal alternatives that could have been deployed without sacrificing performance.

Following the experimental protocol in Chapter 4, the bank trains many models across Logistic Regression, Random Forests, Gradient Boosting, and an MLP, sampling hyperparameters and random seeds. Let $\mathcal{R}_\varepsilon$ be the Rashomon set selected by validation log loss (Chapter 3). For every test applicant i , the audit collects the predictions $(\hat{p}_{mi})_{m \in \mathcal{R}_\varepsilon}$ and computes the observation-wise variance

$$v_i = \text{Var}_{m \in \mathcal{R}_\varepsilon}(\hat{p}_{mi}).$$

Large v_i values flag applicants for whom the predicted risk is highly sensitive to the arbitrary choice of an equally accurate model.

The audit then asks whether unstable cases are isolated outliers or whether they form systematic pockets in feature space. Using standardized numeric features, it builds a simple k -nearest-neighbor graph and checks whether applicants with high v_i tend to appear near each other in feature space. If they do, the bank can treat these as instability hotspots: subpopulations where model-selection uncertainty concentrates and where additional safeguards (e.g., manual review rules, targeted data collection, or reporting prediction uncertainty) may be warranted.

This example illustrates the practical purpose of the methodology developed in this thesis: quantify how much predictions can vary across near-optimal models and identify where this variability concentrates in the feature space.

1.2 Problem Statement

Standard model evaluation workflows select a single best model and report aggregate metrics. This obscures the fact that multiple near-optimal models can imply different outcomes for individual observations. The problem addressed in this thesis is therefore not to find a marginally better model, but to characterize the structure of predictive multiplicity: which observations are unstable, whether instability clusters in feature space, and to what extent instability can be attributed to hyperparameter choices.

Formally, given a hypothesis class $\mathcal{F}$ and a loss function L , we study the set of near-optimal models $\mathcal{R}_\varepsilon$ and analyze the distribution of observation-wise predictive variance $(v_i)_{i=1}^N$ together with its spatial organization under a feature-space neighborhood graph. This yields an observation-level and subgroup-level view of model-selection uncertainty that complements existing aggregate multiplicity metrics.

1.3 Research Questions

Building on the problem described above, this thesis investigates the following research questions:

1. **Observation-wise instability.** How much do predicted probabilities vary across near-optimal models for each observation, and how heterogeneous is this variability across a dataset?
2. **Spatial structure.** Do high-variance observations cluster in feature space (global autocorrelation), and can we localize coherent regions of instability that define unstable subgroups?
3. **Robustness to design choices.** How sensitive are the detected patterns to neighborhood size k , to the tolerance parameter ε , and to the choice of model families included in Rashomon set construction?
4. **Hyperparameter drivers.** Which hyperparameters systematically contribute to predictive variance, and how much instability can be attributed to between-hyperparameter versus within-hyperparameter variation?

Together, these questions aim to provide a structured understanding of how predictive multiplicity manifests across the model, feature, and data spaces, and what this implies for the reliability and transparency of automated decision systems.

1.4 Contributions of This Thesis

1.5 Thesis Structure

2 Background and Related Work

Predictive multiplicity refers to the phenomenon in which several distinct models achieve comparable predictive performance on the same task while differing in their predictions or internal behavior. Recent work shows that such multiplicity is common in modern machine learning pipelines and has implications for fairness, interpretability, and the stability of automated decisions. This section introduces key concepts in the study of predictive multiplicity and summarizes methods proposed to quantify and analyze disagreement across near-optimal models.

2.1 Rashomon Sets in Machine Learning

The concept of a Rashomon set originates from the observation of Breiman (2001) that real-world prediction problems often admit many different models achieving nearly identical accuracy. The term "Rashomon" alludes to Akira Kurosawa's 1950 film *Rashomon*, in which four witnesses provide contradictory yet individually plausible accounts of the same event (a murder in a forest). Just as the film reveals that multiple coherent narratives can coexist without a clear way to determine which is "true," Breiman recognized that machine learning exhibits an analogous phenomenon: for most datasets, "there is often a multitude of different models with essentially equivalent predictive performance".

Breiman termed this the Rashomon effect and argued that it poses a fundamental challenge for model-based inference. When many models fit the data equally well yet differ in their structure, predictions, or implied feature relationships, selecting any single model obscures the existence of these alternatives. This arbitrariness can lead to instability in predictions, misleading interpretations of feature importance, and a systematic underestimation of uncertainty. As a remedy, Breiman advocated methods that embrace rather than ignore this multiplicity—most notably bagging and random forests, which aggregate predictions across many plausible models to produce more stable and reliable outputs.

Modern work has formalized Breiman's intuition by defining the Rashomon set as the collection of all models within a hypothesis class $\mathcal{F}$ whose performance falls within a small tolerance ϵ of the best achievable:

$$\mathcal{R}_\epsilon = \{f \in \mathcal{F} : L(f^*) \leq L(f) + \epsilon\}$$

where $L(f)$ denotes the loss of model f and f^* is the best-performing model. This formalization makes explicit the space of indistinguishably good models that Breiman originally described, enabling systematic study of how predictions, explanations, and fairness properties vary across equally valid modeling choices.

The parameter ϵ controls the tolerance: smaller values yield more restrictive Rashomon sets containing only models very close to optimal, while larger values admit more diverse models. The choice of ϵ reflects a judgment about what constitutes "essentially equivalent" performance, a 1% difference in accuracy may be negligible in some contexts but significant in others. Beyond formal definition, recent work emphasizes that large Rashomon sets can be a practical advantage rather than merely a source of instability. Rudin et al. (2024) argue that multiplicity can be used to satisfy secondary goals (e.g., interpretability or fairness) without sacrificing predictive performance, and Semenova et al. (2022) provide

theoretical support via the Rashomon ratio, showing that large Rashomon sets imply the existence of simpler near-optimal models.

2.2 The Paradigm Shift

Rudin et al. (2024) argue that the existence of large Rashomon sets necessitates a paradigm shift in how we approach predictive modeling. The traditional machine learning workflow—train many models, select the best one, deploy it—treats the selected model as if it were uniquely correct. This obscures the arbitrariness of model selection and the existence of equally valid alternatives.

Instead, Rudin advocates for methods that embrace multiplicity. If the Rashomon set is large, it likely contains models with desirable secondary properties: simpler models that are more interpretable, fairer models that satisfy equity constraints, or models that are more robust to distribution shift. Semenova et al. (2022) formalize this intuition by proving that when the Rashomon set is large (as measured by the Rashomon ratio), there exist simpler models within it that achieve nearly optimal accuracy. This result suggests that interpretability and accuracy need not be in conflict—the conflict arises only when we restrict attention to a single “best” model rather than exploring the full Rashomon set.

2.3 Predictive Multiplicity Metrics

Marx et al. (2020) introduced two foundational metrics that have shaped subsequent research on predictive multiplicity: ambiguity and discrepancy. Both metrics are computed for a baseline classifier (the one with the best performance) on the entire training dataset. Ambiguity is the proportion of points in a training dataset, for which at least one classifier in the Rashomon set predicts a label different from the prediction of the baseline classifier. Discrepancy represents the maximum number of predictions that can change by switching the baseline classifier. High ambiguity indicates that disagreement is common across many points, capturing the overall spread of predictions in the Rashomon set. In contrast, discrepancy reflects a worst-case notion of multiplicity: a high discrepancy means that there exists at least one model in the Rashomon set whose predictions differ substantially from the baseline, even if ambiguity is low overall.

Hsu and Calmon (2022) propose Rashomon Capacity as a measure of the volume of the prediction space covered by the Rashomon set. Rather than examining individual predictions, this metric considers the joint structure of predictions across all samples simultaneously. A high Rashomon Capacity indicates that the Rashomon set contains models that produce diverse patterns of predictions, not just different predictions for a few individuals, but systematically different prediction profiles. This global perspective complements the individual-focused view of ambiguity and discrepancy, capturing the richness of the model space in a single measure.

Cooper et al. (2024) approach multiplicity from a procedural rather than set-based perspective. They define self-consistency as the probability that two independent runs of the same learning algorithm produce models that agree on their prediction for a given sample. This metric does not require explicit construction or enumeration of the Rashomon set;

instead, it estimates multiplicity through repeated training runs. Low self-consistency indicates that predictions are highly sensitive to arbitrary choices in the training procedure (random seeds, weight initialization, data shuffling), even when hyperparameters and model architecture remain fixed.

Hsu and Calmon (2024) introduce the Viable Prediction Range (VPR), which provides a per-sample measure of multiplicity by computing the interval of predictions achievable across all models in the Rashomon set. For each individual, VPR defines the minimum and maximum predicted probability that any near-optimal model assigns to them. The width of this interval quantifies how much uncertainty exists about that individual’s prediction due to model selection alone. VPR makes explicit that different individuals experience different levels of instability: some people receive consistent predictions across all models, while others face substantial uncertainty.

2.4 Limitations of Existing Methods

While these metrics provide valuable perspectives on predictive multiplicity, they share certain limitations that motivate the need for complementary approaches.

- **Aggregate or worst-case summaries.** Ambiguity provides a single number (the worst individual), and discrepancy provides another (the fraction affected). While useful for high-level assessment, these summaries obscure the distribution of instability across individuals. Two datasets with identical ambiguity and discrepancy values could have very different underlying patterns; one might have many individuals with moderate disagreement, while another has few individuals with extreme disagreement. Aggregate metrics do not distinguish these cases.
- **Focus on extremes, not distributions.** Most existing metrics capture the boundaries of the Rashomon set (maximum disagreement, full prediction range) rather than characterizing how predictions are distributed within those boundaries. Two individuals might have identical VPR intervals but experience very different patterns of disagreement: one might have predictions uniformly spread across the interval, while another might have bimodal predictions clustered at the extremes. Range-based metrics treat these cases identically.
- **Binary classification focus.** Discrepancy and self-consistency are defined in terms of classification disagreement; whether models predict different class labels. This binary framing loses information about the magnitude of probability differences. A case where predictions range from 0.49 to 0.51 (crossing the decision boundary) registers the same discrepancy as a case ranging from 0.1 to 0.9, yet these represent fundamentally different types and severities of disagreement.
- **No source attribution.** Existing metrics quantify the amount of multiplicity but do not explain where it comes from. They do not distinguish multiplicity arising from model architecture differences versus hyperparameter choices versus random initialization. Understanding the sources of multiplicity is essential for practitioners who want to mitigate it.

- Limited characterization of prediction uncertainty. While VPR provides per-sample intervals, it does not characterize the shape of the prediction distribution within that interval; whether predictions cluster, spread uniformly, or exhibit multimodality. Such distributional information is crucial for understanding the nature of disagreement.

These limitations motivate our variance-based approach. Rather than summarizing multiplicity with ranges or worst-case values, we compute the variance of predictions across the Rashomon set for each individual. This provides a per-sample metric that captures not just the existence of disagreement but its magnitude and distribution. A variance-based framework also enables decomposition of multiplicity into contributions from different sources (model type, hyperparameters, random seeds), yielding actionable insights for practitioners.

3 Methodology

This chapter presents the methodology used to analyze predictive multiplicity through a spatial lens. The proposed approach characterizes how predictions vary across a set of near-optimal models and investigates whether this variation exhibits statistically significant structure in feature space. The methodology is independent of any particular dataset or experimental configuration; dataset-specific choices are described in Chapter 4.

3.1 Problem Setting and Notation

Let (X, y) denote a supervised learning dataset with N observations, where $X \in \mathbb{R}^{N \times d}$ is the feature matrix and $y \in \{0, 1\}^N$ is a binary target variable. Let $\mathcal{F}$ denote a hypothesis class of predictive models, potentially spanning multiple model families and hyperparameter configurations.

We consider probabilistic classifiers $f \in \mathcal{F}$ that output predicted probabilities $\hat{p}_{fi} \in [0, 1]$ for each observation $i \in \{1, \dots, N\}$. Model performance is measured using a loss function $L(f)$ evaluated on held-out data.

3.2 Rashomon Set of Near-Optimal Models

Predictive multiplicity is studied through the Rashomon set, defined as the collection of models whose performance lies within a tolerance $\varepsilon > 0$ of the best achievable loss. Formally, let

$$L^* = \min_{f \in \mathcal{F}} L(f)$$

denote the minimal loss over all trained models. The Rashomon set is then defined as

$$\mathcal{R}_\varepsilon = \{f \in \mathcal{F} : L(f) \leq L^* + \varepsilon\}.$$

The tolerance parameter ε controls the strictness of the selection: smaller values yield a more homogeneous set of models, while larger values admit greater diversity. The Rashomon set serves as the basis for all subsequent analyses.

3.3 Observation-Wise Predictive Variance

To quantify predictive multiplicity at the level of individual observations, we compute the variance of predicted probabilities across the Rashomon set. Let $|\mathcal{R}_\varepsilon| = M$ denote the number of models in the Rashomon set, and let $\hat{p}_{mi}$ denote the predicted probability for observation i produced by model $m \in \mathcal{R}_\varepsilon$.

The observation-wise predictive variance is defined as

$$v_i = \text{Var}_{m \in \mathcal{R}_\varepsilon}(\hat{p}_{mi}), \quad i = 1, \dots, N.$$

This yields a scalar instability score for each observation, capturing the magnitude of disagreement among near-optimal models. Unlike range-based or binary disagreement metrics, predictive variance reflects the full distribution of predictions within the Rashomon set.

3.4 Feature-Space Graph Construction

To study the spatial organization of predictive variance, we represent the dataset as a graph in feature space. Observations are treated as nodes, and edges encode local neighborhood relationships.

A k -nearest-neighbor (kNN) graph is constructed using the feature vectors in X . Each observation is connected to its k nearest neighbors according to a chosen distance metric. The resulting adjacency structure is represented by a row-normalized spatial weight matrix $W \in \mathbb{R}^{N \times N}$, where $W_{ij} > 0$ indicates that observation j is a neighbor of observation i .

This graph provides the spatial structure required for subsequent autocorrelation and clustering analyses.

3.5 Global Spatial Autocorrelation

To test whether predictive variance is spatially structured in feature space, we compute the global Moran's I statistic. Let $\mathbf{v} = (v_1, \dots, v_N)^\top$ denote the vector of observation-wise predictive variances, and let $\bar{v}$ denote its mean.

Global Moran's I is defined as

$$I = \frac{\sum_{i=1}^N \sum_{j=1}^N W_{ij} (v_i - \bar{v})(v_j - \bar{v})}{\sum_{i=1}^N (v_i - \bar{v})^2}.$$

Positive values of I indicate that observations with similar levels of predictive variance tend to be neighbors in feature space, while values near zero suggest spatial randomness. Statistical significance is assessed using permutation tests, in which the variance values are randomly permuted across observations while the spatial structure encoded by W is held fixed.

3.6 Local Indicators of Spatial Association

While global Moran's I assesses overall spatial dependence, it does not identify where such dependence occurs. To localize regions of elevated predictive multiplicity, we compute Local Indicators of Spatial Association (LISA).

For each observation i , the local Moran statistic is defined as

$$I_i = (v_i - \bar{v}) \sum_{j=1}^N W_{ij} (v_j - \bar{v}).$$

Permutation-based significance tests are performed independently for each observation. Based on the sign of $v_i - \bar{v}$ and the local spatial lag, observations are assigned to one of four categories: High-High (HH), Low-Low (LL), High-Low (HL), or Low-High (LH). Observations that do not reach statistical significance are labeled as non-significant. High-High observations correspond to locally unstable regions in which high predictive variance is surrounded by similarly unstable neighbors.

3.7 Connected Components of High-Multiplicity Regions

To extract coherent regions of predictive multiplicity, we restrict the kNN graph to observations classified as High–High and identify connected components within this subgraph. Each connected component represents a contiguous region of feature space in which predictive variance is consistently elevated. Components below a minimum size threshold are discarded to avoid spurious detections.

This component-based representation enables region-level analysis and supports downstream interpretability and stability assessments.

3.8 Null Models for Spatial Multiplicity

To ensure that detected spatial structure is not an artifact of marginal variance distributions, we employ null models based on permutation of predictions.

Specifically, predictions are permuted independently within each model across observations, preserving the overall distribution of predictions for each model while destroying any alignment between predictive variance and feature space. Observation-wise variance and spatial statistics are recomputed on the permuted data.

Comparing observed statistics to their null distributions allows us to assess whether spatial clustering of predictive variance exceeds what would be expected under spatial randomness.

3.9 Hyperparameter Variance Decomposition

To investigate the sources of predictive multiplicity, we decompose observation-wise variance into contributions attributable to hyperparameter choices.

For a given hyperparameter h , models in the Rashomon set are grouped according to their value of h . For each observation, total variance is decomposed into between-group and within-group components:

$$\text{Var}_{\text{total}} = \text{Var}_{\text{between}}(h) + \text{Var}_{\text{within}}(h).$$

This decomposition quantifies how much of the predictive instability for each observation is explained by differences in hyperparameter values versus residual variation within hyperparameter groups. Aggregating these quantities across observations provides a principled measure of hyperparameter importance for predictive multiplicity.

3.10 Summary

The methodology described in this chapter provides a unified framework for analyzing predictive multiplicity as a spatial phenomenon. By combining Rashomon set analysis, observation-wise predictive variance, spatial statistics, null models, and hyperparameter decomposition, the approach enables identification, localization, and interpretation of regions in feature space where model predictions are particularly unstable.

4 Implementation Details

This section describes the concrete implementation of the experimental pipeline used in this thesis. It summarizes the inputs, model training procedure, Rashomon set construction, cross-validation options, and the artifacts produced for downstream analysis.

4.1 Inputs

The pipeline takes as input a dataset (X, y) with binary target y , feature metadata (numeric vs. categorical), a tolerance parameter ε for Rashomon set selection, and a set of model families with associated hyperparameter search spaces. Experiments additionally specify the number of sampled configurations per family (n_{models}) and one or more random seeds.

4.2 Model families and preprocessing

We train models across four families: Logistic Regression, Random Forests, Gradient Boosting, and a multilayer perceptron (MLP). Numeric features are standardized for Logistic Regression and the MLP, while categorical features are one-hot encoded for all model families. For each family, hyperparameter configurations are sampled uniformly at random from a pre-defined search space and trained with one or more seeds to account for stochasticity.

4.3 Rashomon set construction

A trained model m is included in the Rashomon set if its validation loss satisfies $L(m) \leq L^* + \varepsilon$, where L^* is the best observed validation loss across all trained models. We support two selection modes: a global threshold applied across all families, and a per-family threshold that uses the best loss within each family as the reference.

4.4 Cross-validation and test set usage

To reduce dependence on a single split, the pipeline can compute loss using either K -fold cross-validation (default $K = 5$) or repeated holdout splits. A separate test set is held out from all training and selection decisions and is used only to generate final predictions for models in the Rashomon set and to compute observation-wise multiplicity quantities.

4.5 Saved artifacts

For each run, we save: (i) a predictions matrix $P_{\text{test}} \in \mathbb{R}^{M \times N}$ for all Rashomon models on the test set, (ii) a metadata table describing each model and its hyperparameters, (iii) per-observation multiplicity metrics derived from P_{test} , and (iv) the split information and configuration used to reproduce the experiment.

5 Experimental setup

This chapter describes how the methodology introduced in Chapter 3 is instantiated in empirical experiments. We specify the datasets, model families, training protocol, Rashomon set construction, and spatial analysis parameters used throughout the thesis. All design choices are fixed across experiments unless stated otherwise.

5.1 Datasets

Experiments are conducted on three benchmark datasets commonly used in the study of fairness, interpretability, and predictive uncertainty:

- **COMPAS**: A criminal recidivism dataset containing demographic information, criminal history, and a binary outcome indicating whether an individual reoffended within two years.
- **German Credit**: A credit risk dataset consisting of financial and demographic attributes, with a binary label indicating good or bad credit risk.
- **Breast Cancer Wisconsin**: A medical dataset containing numeric features derived from cell nuclei measurements, with a binary malignancy label.

All datasets involve binary classification tasks and include a mix of numeric and categorical features. Dataset-specific preprocessing steps are described below.

5.2 Data Splitting

Each dataset is split into disjoint training, validation, and test sets. The test set is held out from all model training, hyperparameter selection, and Rashomon set construction procedures.

Unless otherwise stated, splits are created using a fixed random seed to ensure reproducibility. Stratified sampling is used when applicable to preserve class proportions across splits. The test set is used exclusively for:

- generating predictions from Rashomon models,
- computing observation-wise predictive variance,
- performing spatial autocorrelation and clustering analyses.

No information from the test set is used during model fitting or Rashomon selection.

5.3 Model Families

To induce predictive multiplicity, models are trained across four commonly used model families with diverse inductive biases:

- **Logistic Regression (LogReg)**

- **Random Forest (RF)**
- **Gradient Boosting Machines (GBM)**
- **Multilayer Perceptron (MLP)**

These families span linear, tree-based, ensemble, and neural network models, enabling analysis of multiplicity arising from both model architecture and hyperparameter variation.

5.4 Hyperparameter Sampling

For each model family, hyperparameter configurations are sampled uniformly at random from predefined search spaces. The search spaces are chosen to cover commonly used and practically relevant settings, including regularization strength, tree depth, ensemble size, learning rate, and network architecture.

For each model family, a fixed number of hyperparameter configurations is sampled. Each configuration is trained with one or more random seeds when the model family exhibits stochasticity (e.g., Random Forests, MLPs). This induces additional variability within the Rashomon set and allows separation of hyperparameter-driven and seed-driven effects.

5.5 Preprocessing

Numeric features are imputed using the median and standardized to zero mean and unit variance when required by the model family. Categorical features are imputed using the most frequent category and encoded using one-hot encoding.

Preprocessing is implemented using a pipeline and is fit exclusively on the training data. The same preprocessing transformations are applied to validation and test data without refitting.

5.6 Model Evaluation and Rashomon Set Construction

Model performance is evaluated using log loss computed on validation data or, when cross-validation is enabled, using mean validation loss across folds.

Given a tolerance parameter ε , the Rashomon set is constructed as described in Chapter 3 by selecting all models whose validation loss lies within ε of the best observed loss. Two selection modes are considered:

- **Global:** a single threshold applied across all model families,
- **Per-family:** a separate threshold applied within each model family.

Unless otherwise stated, the global selection mode is used. The tolerance ε is fixed across datasets to ensure comparability of results.

5.7 Prediction Matrix Construction

For each dataset, all models in the Rashomon set are applied to the held-out test set to obtain predicted probabilities. This yields a prediction matrix

$$P \in \mathbb{R}^{M \times N},$$

where M is the number of Rashomon models and N is the number of test observations. Observation-wise predictive variance is computed from this matrix as described in Section 3.3.

5.8 Feature-Space Graph Parameters

Spatial analyses are conducted using a k -nearest-neighbor graph constructed in feature space. The graph is built using standardized numeric features. Only numeric features are used to define distances to ensure a well-defined metric structure.

Unless stated otherwise, the neighborhood size k is fixed across datasets. Sensitivity analyses over different values of k are conducted to assess robustness of spatial autocorrelation results.

The resulting adjacency matrix is row-normalized and used as the spatial weight matrix for all Moran's I and LISA computations.

5.9 Spatial Autocorrelation and Clustering

Global spatial autocorrelation is assessed using Moran's I with permutation-based significance testing. Local spatial clustering is analyzed using Local Indicators of Spatial Association (LISA), with statistical significance assessed via permutation tests and multiple-testing correction.

Observations classified as High-High (HH) are interpreted as belonging to locally unstable regions. Connected components of HH observations are extracted from the restricted kNN graph, with a minimum component size threshold applied to avoid spurious detections.

5.10 Null Model Configuration

Null models are constructed by permuting predictions independently within each model across observations. This preserves marginal prediction distributions while destroying any alignment between predictive variance and feature space.

For each null run, observation-wise variance and spatial statistics are recomputed. Repeating this procedure yields empirical null distributions for Moran's I and HH cluster counts, against which observed values are compared.

5.11 Hyperparameter Analysis

To analyze the sources of predictive multiplicity, hyperparameter variance decomposition is performed within selected model families. Observation-wise predictive variance is decomposed into between-hyperparameter and within-hyperparameter components as described in Section 3.9.

This analysis is restricted to models belonging to the Rashomon set and is used to identify hyperparameters that systematically contribute to prediction instability.

5.12 Implementation Details

All experiments are implemented in Python using `scikit-learn` for model training and evaluation, and `NumPy` and `SciPy` for numerical computations. Spatial statistics and permutation tests are implemented explicitly to ensure full control over the analysis pipeline. Random seeds are fixed throughout to ensure reproducibility. All code, configuration files, and generated results are provided in the electronic appendix.

6 Results

This chapter reports the empirical results of the analysis. We focus on (i) the size and composition of the Rashomon set, (ii) the distribution of observation-wise predictive variance, (iii) whether instability clusters in feature space, and (iv) how much instability can be attributed to model and hyperparameter choices. Implementation details of the pipeline are described in Chapter 4.

6.1 Rashomon set composition

We begin by summarizing how many trained models fall into the Rashomon set under the chosen tolerance ε , and how these models are distributed across model families. This provides a basic check that multiplicity is present and that the Rashomon set contains meaningful diversity rather than minor perturbations of a single model type.

When reporting composition, we recommend including at least:

- the total number of trained models and the Rashomon set size $|\mathcal{R}_\varepsilon|$,
- the per-family counts in $\mathcal{R}_\varepsilon$,
- the best (minimum) and worst (maximum) loss among Rashomon models.

6.2 Observation-wise predictive variance

For each test observation i , we compute the variance of predicted probabilities across Rashomon models,

$$v_i = \text{Var}_{m \in \mathcal{R}_\varepsilon}(\hat{p}_{mi}).$$

We summarize the distribution of $(v_i)_{i=1}^N$ using quantiles (e.g., median, 90th, 95th, 99th percentile) and visualization. In particular, we are interested in whether instability is concentrated in a small subset of observations (a heavy right tail) or spread more uniformly. To make results comparable across datasets, it can be helpful to report variance together with complementary per-observation summaries, such as prediction interval width or flip instability at a fixed threshold.

6.3 Spatial structure of instability

We then test whether high-variance observations tend to occur near each other in feature space using the neighborhood graph described in Chapter 3. If instability is spatially structured, we summarize the resulting high-instability pockets and describe their characteristic feature profiles (e.g., via group-wise summary statistics).

6.4 Drivers of predictive multiplicity

Finally, we analyze which design choices contribute most to predictive variance. We compare instability patterns across model families and evaluate hyperparameter-level effects using variance decomposition. This helps distinguish instability driven primarily by model class choice from instability that persists even within a fixed family.

7 Discussion

8 Conclusion

A concise summary of contents and results

A Appendix

Additional material goes here

B Electronic appendix

Data, code and figures are provided in electronic form.

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